

marks)

- 4 (a)** field causes forces on the electronsM1
and the nucleus in opposite directions A1
(field causes) electrons (to be) stripped off the atom..... B1 [3]
- (b) (i)** $E = Q/4\pi\epsilon_0 r^2$ C1
 $20 \times 10^3 \times 10^2 = Q/(4\pi \times 8.85 \times 10^{-12} \times 0.21^2)$ C1
charge = 9.8×10^{-6} CA1 [3]

(ii) $V = Q/4\pi\epsilon_0 r$
 $= (9.8 \times 10^{-6})/(4\pi \times 8.85 \times 10^{-12} \times 0.21) \dots\dots\dots C1$
 $= 4.2 \times 10^5 \text{ V} \dots\dots\dots A1$ [2]

(c) e.g. sphere not smooth, humid air, etc B1 [1]

5 (a) centripetal force = mv^2/r B1